



CookBookCommunity

A Recipe Sharing Application

Group 04 — Even Group

Jetpack Compose

Firebase Firestore

MVI Architecture

Product & Strategy



Core Concept

A community-first recipe platform where **anyone can explore** recipes, but only logged-in users can **publish, review, and save** them.

Our Differentiators vs. Base Lab1

- Recipe adaptation (fork & personalize)
- Cooking diary / tried-recipe log
- Customizable favorite collections

Strategic Vision



Social Discovery

Users inspire each other by forking and adapting recipes publicly.



Personal Archive

Diary entries & favorite collections make the app a lasting cookbook.



Engagement Loop

Notifications close the loop: review → author notified → community grows.

Main Targets — Personas

Giulia, 27 — Turin

Working student, little time to cook

- Quick filters & easy-to-follow recipes
- Clear time, cost & difficulty info
- Tips from people who've tried it
- Saves recipes into named collections

Marco, 33 — Rome

Shares his own recipe twists online

- Easy-to-use recipe publishing form
- Notified the moment someone reviews
- Gets credited when recipes are adapted
- Consults reviews on his own recipes

Peppina, 69 — Palermo

Cooking vegan meals for her grandson

- Filters recipes by dietary restriction
- Clearly listed ingredients, always
- Tips & suggested substitutions
- Notified of recipes matching her profile

Key Elements & Requirements



Recipe CRUD

Create, edit, delete recipes with image, steps, ingredients, time, difficulty, price & portions.



Browse & Filter

Filter by type, difficulty, time, price & diet — available without login.



Auth + Profiles

Google Sign-In via Credential Manager. Rich profile: cooking role, bio, preferences, social links, avatar/monogram.



Save to Collections

Bookmark recipes into named collections (Saved / Tried + user-created). Atomic Firestore batch writes.



Reviews & Tips

Star-rated reviews with photo and title; quick text tips. Denormalised rating avg on recipe doc via batch.



Navigation Guards

Unauthenticated users can browse freely; actions like save or publish redirect to login with post-login destination.

Extra Features



Recipe Adaptation

Fork any recipe, add adaptation notes — original author is credited & notified.



Cooking Diary

Log each cook attempt: result, photo, modifications, would-cook-again flag.



Custom Collections

Create named favourite lists; save recipes into any collection.

Extra Libraries & Tech Stack

Image Loading

Coil Compose

Async image loading & caching for Jetpack Compose. Used for recipe and profile photos throughout the app.

Authentication

Firebase Auth

Google Sign-In via Credential Manager. Manages sign-up flow, session persistence, and auth state.

Database

Firebase Firestore

Real-time NoSQL backend. Snapshot Flows for live updates; atomic batch writes for denormalised data.

Serialization

Kotlinx Serialization

Type-safe JSON serialization used with Compose Navigation to pass complex route arguments.

Auth Integration

Credentials Play Services

Google Identity Services integration for one-tap Google Sign-In on Android.

Async / Flows

Kotlinx Coroutines

StateFlow, Flow, and viewModelScope power all async data operations and UI state management.